American Computer Science League

2025-26 Finals ● Short Problems Solutions ● Intermediate/Classroom Division

1. Boolean Algebra

$\overline{\overline{A + B} * C + \overline{A}(B + C) + \overline{AC} * B}$
$= \overline{A} * \overline{B} * C + \overline{A}B + \overline{A}C + (\overline{A} + \overline{C})B$
$= \overline{A} * \overline{B} * C + \overline{A}B + \overline{A}C + \overline{A}B + \overline{C}B$
$\overline{A} * \overline{B} * C + \overline{A}B + \overline{A}C + B\overline{C}$
This is TRUE for (0,0,1), (0,1,*), (0,*,1), (*,1,0) which are 001, 010, 011, 110 so there are 4 of them.

C. 4

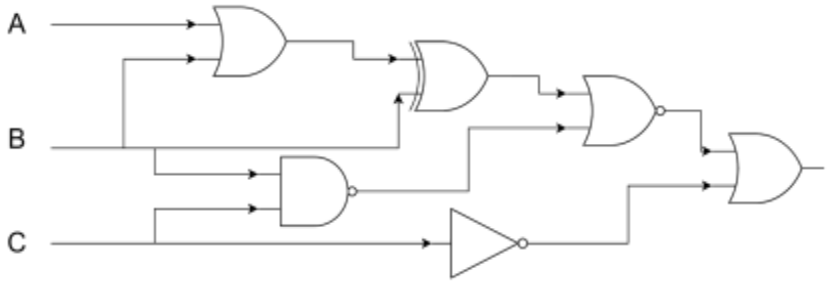
2. Boolean Algebra

$(A + \overline{B})(\overline{\overline{B} + \overline{C}}) + \overline{(A + \overline{B\,C} + C(\overline{\overline{A} + B}))}$
$= (A + \overline{B})BC + \overline{A} * BC * \overline{CA\overline{B}}$
$= ABC + \overline{B}BC + \overline{A}BC * (\overline{C} + \overline{A} + B)$
$= ABC + \overline{A}BC\overline{C} + \overline{A}BC\overline{A} + \overline{A}BCB$
$= ABC + \overline{A}BC + \overline{A}BC$
$= BC(A + \overline{A})$
$= BC$

B. BC

3. Digital Electronics

The digital circuit



is translated to the Boolean algebra expression:
$\overline{((A + B) \oplus B) + \overline{\overline{BC}} + \overline{C}}$

A. $\overline{((A + B) \oplus B) + \overline{\overline{BC}} + \overline{C}}$

4. Bit-String Flicking

(LCIRC-2 $A5_{16}$) OR (RSHIFT-1 10010101_2)
= (LCIRC-2 10100101) OR 01001010
= 10010110 OR 01001010
= 11011110
= 336_8

D. 336

5. Bit-String Flicking

(LCIRC-3 ((RSHIFT-1 01111) OR (RCIRC-2 X) AND (LSHIFT-1 11111)))
= (LCIRC-3 (00111 OR deabc AND 11110))
= (LCIRC-3 (00111 OR deab0)) =
= (LCIRC-3 de111)
= 11de1
11de1 = 11001 makes d=0 and e=0 so the string abcde = ***00.

B. ***00

6. Recursive Functions

If

$$f(x) = \begin{cases} 2x+1 & \text{if } x < 3 \\ 2 \cdot f(x-2) - f(x-1) + 3 & \text{if } x \geq 3 \end{cases}$$

$f(7) = 2 * f(5) - f(6) + 3 = 2 * 2 - f(6) + 3$
$= 4 - 19 + 3 = -12$
$f(5) = 2 * f(3) - f(4) + 3 = 2 * 4 - 9 + 3 = 2$
$f(3) = 2 * f(1) - f(2) + 3 = 2 * 3 - 5 + 3 = 4$
$f(1) = 2 * 1 + 1 = 3, f(2) = 2 * 2 + 1 = 5$
$f(4) = 2 * f(2) - f(3) + 3 = 2 * 5 - 4 + 3 = 9$
$f(6) = 2 * f(4) - f(5) + 3 = 2 * 9 - 2 + 3 = 19$

A. -12

7. Recursive Functions

If

$$f(x.y) = \begin{cases} f(y-3, x+1) - 3 & \text{if } x + y \geq 8 \\ 2 \cdot f(x-2, y+1) + 2 & \text{if } 5 \leq x + y < 8 \\ x + y & \text{if } x + y < 5 \end{cases}$$

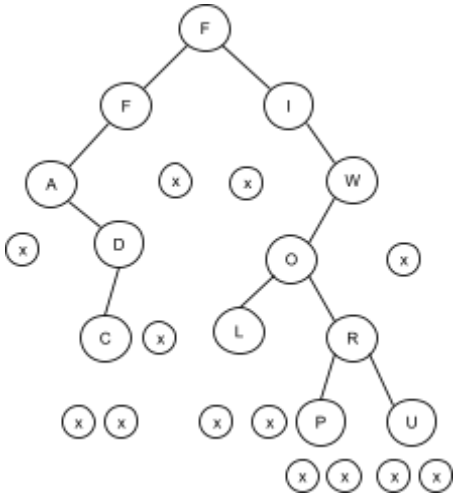
$f(5,7) = f(4,6) - 3 = 13$
$f(4,6) = f(3,5) - 3 = 16$
$f(3,5) = f(2,4) - 3 = 19$
$f(2,4) = 2 * f(0,5) + 2 = 22$
$f(0,5) = 2 * f(-2,6) + 2 = 10$
$f(-2,6) = 4$

D. 13

8. Prefix/Infix/Postfix Notation If a & b = $a^2 - b$ and a % b = [(a+b) / 2] where [x] is the greatest integer less than or equal to x, then 4 3 % 3 3 & 2 1 & * 3 2 ^ / + 8 2 3 % / − = ((4 % 3) + (((3 & 3) * (2 & 1) / (3 ^ 2))) - (8 / (2 % 3)) = (3 + 6 * 3 / 9) - 8 / 2 = 3 + 18 / 9 - 4 = 1	B. 1
9. Prefix/Infix/Postfix Notation $\frac{(A - B^3 + C)^2}{(\frac{A+C}{B})}$ = (((A − (B ^ 3)) + C) ^ 2) / ((A + C) / B) = (((A − (^ B 3) + C) ^ 2) / ((+ A C) / B) = ((− A ^ B 3) + C) ^ 2) / (/ + A C B) = ((+ − A ^ B 3 C) ^ 2) / (/ + A C B) = (^ + − A ^ B 3 C 2) / (/ + A C B) = / ^ + − A ^ B 3 C 2 / + A C B	A. /^+−A^B3C2/+ACB
10. Computer Number Systems The average of 152_8, 271_8, and 713_8 can be found by adding the 3 octal numbers in base 8, then converting the sum to base 10, and finally dividing that answer by 3. $(152_8+271_8+713_8) / 3_8 = 1356_8 / 3_8 = 750_{10} / 3_{10} = 250_{10}$ 1356 = 1*512+3*64+5*8+6 = 512+192+46 = 750	A. 250
11. Computer Number Systems To evaluate $(ABC_{16} + 864_{16}) / 10_{16}$ add in base 16 and remove the last 0 when dividing. Then convert from hexadecimal to octal using binary. $(ABC_{16} + 864_{16}) / 10_{16}$ $= 1320_{16} / 10_{16} = 132_{16}$ $132_{16} = 100\ 110\ 010_2 = 462_8$	B. 462

12. Data Structures

The binary search tree for FIFAWORLDCUP with NIL nodes is:



The external path length is the sum of the depths of the NIL nodes:
2*2+2*3+4+4*5+4*6=58

C. 58

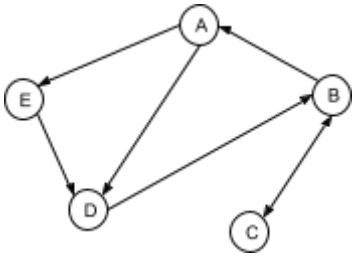
13. Data Structures

The following Undo and Redo commands
↩ ↩ ↩ ↩ ↩ ↩ ↪ ↩ ↩ ↪ ↩ ↩ ↪ ↪ ↪ ↩ ↩ ↩ ↩ ↩
result in 'ACSL p' after the first 6 ↩, adds 'r', removes r and p, adds p, removes p and <sp>, adds <sp>, 'p', and 'r', removes 'r', 'p', <sp>, 'L', and 'S'. You still have AC.

D. AC

14. Graph Theory

The adjacency matrix for the graph given must be cubed to get paths of length 3 as follows:



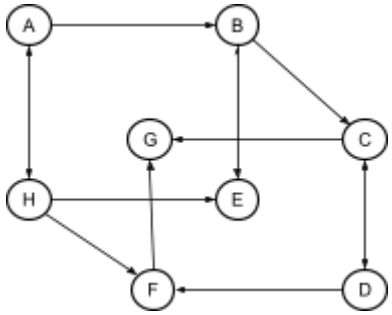
$$\begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}^3 = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

E has 2; A, C, and D have 3; and B has 4.

B. B

15. Graph Theory

In this graph the only cycles are: AHA, CDC, BEB. There are none with 3 or 4 and no others that connect the squares. Therefore, there are 0 cycles with more than 2 different vertices. That answer is not given so the answer is E (none of the above).



E. None of the above

16. What Does This Program Do?

```
for j = 0 to len(arr)-1
   min = j
   for k = j + 1 to len(arr)-1
      if arr(k)<arr(min) then
         min = k        .
      end if
   next k
   t = arr(j)
   arr(j) = arr(min)
   arr(min) = t
next j
```

This program compares each item but the last with the numbers after it in the array. Each time it swaps the next smallest with that number if necessary.

C. `min = k`

17. LISP Programming

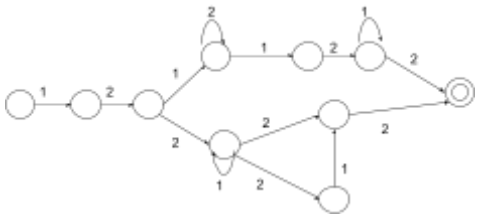
(SETQ X '((SUB 7 2) 5 (ADD 3 1)))
(SETQ Y '((MULT 4 3 5) (DIV 2 1) (6 (3 4) 2)))
(EXP (CADDR X) (CADR Y))
This evaluates to (EXP (ADD 3 1) (DIV 2 1)) = 4^2 = 16

C. 16

18. FSAs and Regular Expressions

Given this FSA at the right:

a. 12122222
   fails on bottom at 4th 2
b. 1222112
   fails on bottom at 2nd 1
c. 121122 - OK on top branch
d. 1221111111212 -
   OK on bottom branch
e. 12122222122 - OK on top branch
f. 122111212 - OK on bottom branch



B. c,d,e,f

19. FSAs and Regular Expressions

Given the regular expression: [A-C] [a-hj-np-z]* . [1-9] [0-9]*,

Bfractals#8000 is OK	Afib:11235 fails at i
Carfun@1089 is OK	Calculate!0456 fails at 0
Addsum+6789 is OK	Bravado012 fails at 0

There are 3 of them that are accepted.

B. 3

20. Assembly Language

```
A       DC      13
B       DC      21
NUM     DC      7
TOP     LOAD    NUM
        SUB     =11
        BE      STOP
        ADD     =12
        STORE   NUM
        LOAD    A
        ADD     B
        STORE   C
        LOAD    B
        STORE   A
        LOAD    C
        STORE   B
        BU      TOP
STOP    PRINT   B
        END
```

This program outputs the 12th Fibonacci number in the sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 starting with 21 as the 7th term.

It is the only Fibonacci number that is also the square of its term number since $12^2 = 144$.

D. 144